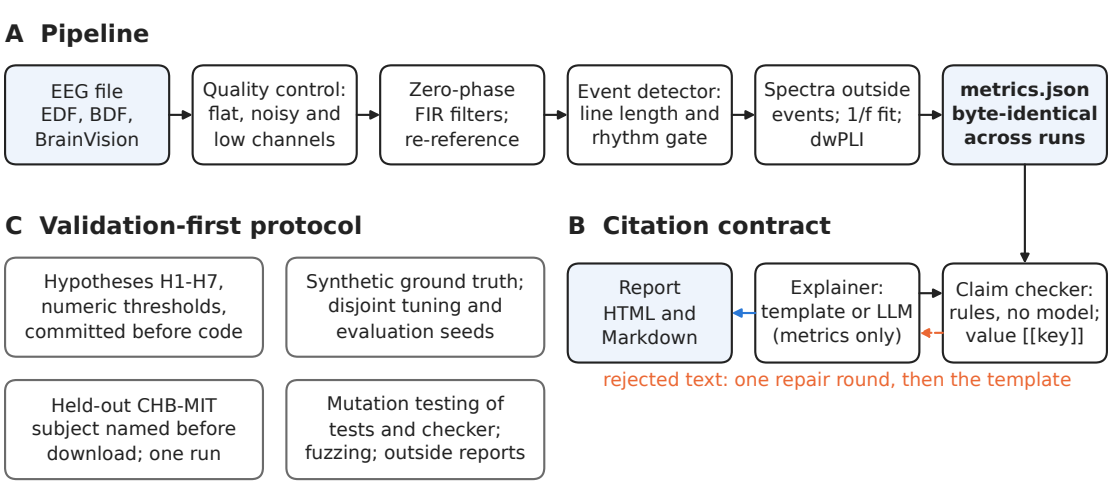
A Pipeline
EEG file EDF, BDF, BrainVision
Quality control: flat, noisy and low channels
Zero-phase FIR filters; re-reference
Event detector: line length and rhythm gate
Spectra outside events; 1/f fit; dwPLI
metrics.json byte-identical across runs
B Citation contract
Claim checker: rules, no model; value [[key]]
Explainer: template or LLM (metrics only)
Report HTML and Markdown
rejected text: one repair round, then the template
C Validation-first protocol
Hypotheses H1-H7, numeric thresholds, committed before code
Synthetic ground truth; disjoint tuning and evaluation seeds
Held-out CHB-MIT subject named before download; one run
Mutation testing of tests and checker; fuzzing; outside reports